

Idea Abstract

TEAM RUBIXCUBE · SIH 2026 · PS 26097

Problem Statement 26097, Ministry of Social Justice and Empowerment. Team RUBIX CUBE, Team ID 140224. Solution name: Kaushal Saathi.

Abstract

Many Scheduled Caste beneficiaries under the Grant in Aid component of PM-AJAY struggle with text heavy digital processes. Low digital literacy, language barriers and long forms mean a person often needs someone else's help to apply, or gives up midway. The result is a mismatch between the beneficiary, the training she enrolls in and the work actually available near her. The CAG audit of PMKVY in 2025 found that about 41 percent of certified candidates under its short term components were placed, which shows that certification by itself does not lead to employment.

Kaushal Saathi is a multilingual voice first assistant that replaces the form with a conversation. The beneficiary speaks in her own language over a normal phone call through IVR, over a WhatsApp voice note, or through a light mobile and kiosk app. The system asks short questions about her education, family occupation, current work, skills, interests, constraints and preference for self employment or a job. It converts the conversation into a structured livelihood profile and reads the answers back for confirmation.

Before recommending anything, the system checks her profile against the published entry requirements of NSQF aligned qualifications taken from the National Qualification Register. She is shown only the pathways she qualifies for, along with the exact requirement that is missing for the ones she does not. Recommendations are matched to opportunities near her, including vacancies posted by verified local employers in her district, so training points towards real work. Follow up calls and messages keep her informed about training dates and next steps.

The same records give district and state officers a dashboard of demand: which trades people are asking for, which skill gaps repeat, and what employers need. This supports perspective planning, beneficiary monitoring, coordination and outcome tracking under PM-AJAY.

Speech is handled through Bhashini and Sarvam, telephony through IVR and WhatsApp, and data is stored with role based access. Only the information needed for eligibility and mapping is collected, audio is discarded after the transcript is confirmed, and the design follows the Digital Personal Data Protection Act, 2023.

What makes it different

We check eligibility before we recommend, and we name the missing requirement instead of hiding it.

One interview engine works across IVR, WhatsApp and the app, so a feature phone user is not left out.

Verified local employers post real vacancies, which connects skilling to employment in the same district.

The beneficiary side and the officer side share one record, so planning uses live demand instead of periodic manual collection.